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Electric machine and method for operating said machine

Abstract

An electric machine, including a rotor having a rotor shaft and a rotor laminated core, a stator having a stator laminated core with recesses and stator windings disposed in the recesses of the stator laminated core, wherein coolant can flow through the recesses such that the stator windings can be directly cooled by the coolant, wherein channels are configured in the stator laminated core at a distance to the recesses such that the stator windings can be indirectly cooled by the coolant flowing through the channels, and a device via which a flow of coolant through the recesses and through the channels can be set depending on an operating point of the electric machine such that, at first operating points, the coolant can be conducted only through the recesses and, at second operating points, the coolant can be conducted at least through the channels.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims benefit to German Patent Application No. DE 10 2021 130 471.7, filed on Nov. 22, 2021, which is hereby incorporated by reference herein.

FIELD

(2) The invention relates to an electric machine. The invention also relates to a method for operating an electric machine.

BACKGROUND

(3) The basic design of an electric machine is known from practice. An electric machine therefore

comprises a stator, which is also referred to as the stand, and a rotor, which is also referred to as the runner.

(4) DE 10 2017 102 141 A1 discloses an electric machine comprising a rotor and a stator. Between the rotor and the stator there is an air gap. Recesses in which stator windings are accommodated are configured in the stator. The stator windings can be cooled via a coolant that flows directly around the stator windings. To prevent the coolant from flowing from the stator in the direction of the rotor via the air gap formed between the stator and the rotor, a sleeve-like sealing means is disposed between the stator and the rotor.

(5) DE 10 2019 131 069 A1 discloses another electric machine comprising a rotor and a stator. Recesses that accommodate stator windings are again configured in the stator laminated core of the stator. Channels, through which coolant can flow, are configured in the stator laminated core at a distance to said recesses which accommodate the stator windings. The coolant flowing through these channels does not flow directly around the stator windings. This coolant flowing through the channels rather causes indirect cooling of the stator windings. Between two recesses which accommodate the stator windings and which are adjacent to one another in peripheral direction, there is at least one cooling channel through which coolant flows, which, viewed in peripheral direction, has a defined distance to the recesses which accommodate the stator windings.

(6) DE 10 2011 056 007 A1 discloses another electric machine comprising a rotor and a stator. The stator comprises a stator laminated core with recesses in which stator windings are again disposed. Channels for coolant are moreover configured in the stator laminated core at a distance to said recesses which accommodate the stator windings, specifically at a radial distance.

(7) DE 10 2016 222 331 A1 discloses an electric machine comprising a stator and a rotor, wherein, in addition to the recesses that serve to accommodate stator windings, channels for coolant are configured in the stator laminated core at a distance to said recesses.

(8) DE 10 2005 000 642 A1 and WO 2019 238 365 A1 disclose other electric machines.

(9) There is a need for an electric machine with improved cooling of the stator. There is in particular a need to increase the continuous power density of the electric machine through improved cooling of the stator of the electric machine.

SUMMARY

(10) In an embodiment, the present disclosure provides an electric machine, comprising a rotor comprising a rotor shaft and a rotor laminated core, a stator comprising a stator laminated core with recesses and stator windings disposed in the recesses of the stator laminated core, wherein coolant can flow through the recesses which accommodate the stator windings in such a way that the stator windings can be directly cooled by the coolant, wherein channels are configured in the stator laminated core at a distance to the recesses which accommodate the stator windings through which coolant can flow in such a way that the stator windings can be indirectly cooled by the coolant flowing through the channels, and a device via which a flow of coolant through the recesses which accommodate the stator windings and through the channels can be set depending on an operating point of the electric machine in such a way that, at first operating points, the coolant can be conducted only through the recesses which accommodate the stator windings and, at second operating points, the coolant can be conducted at least through the channels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached

drawings, which illustrate the following:

(2) FIG. 1 shows a cross-section through a first electric machine in the axial direction of the section;

(3) FIG. 2 shows the cross-section II-II of FIG. 1;

(4) FIG. 3 shows the cross-section III-III of FIG. 1; and

(5) FIG. 4 shows a cross-section through a second electric machine in the axial direction of the section.

DETAILED DESCRIPTION

(6) In an embodiment, the present invention provides a novel electric machine with improved cooling of the stator and a method for operating the electric machine.

(7) The electric machine according an embodiment of the invention has a rotor comprising a rotor shaft and a rotor laminated core and a stator comprising a stator laminated core with recesses and stator windings disposed in the recesses of the stator laminated core.

(8) Coolant can flow through the recesses which accommodate the stator windings in such a way that the stator windings can be directly cooled by the coolant flowing through said recesses.

(9) Channels are configured in the stator laminated core at a distance to the recesses which accommodate the stator windings through which coolant can flow in such a way that the stator windings can be indirectly cooled by the coolant flowing through said channels.

(10) The electric machine according to an embodiment of the invention further comprises a device via which a flow of coolant through the recesses which accommodate the stator windings and through the channels can be set depending on the operating point of the electric machine in such a way that, at first operating points, the coolant can be conducted only through the recesses which accommodate the stator windings and, at second operating points, the coolant can be conducted at least through the channels.

(11) The electric machine according to an embodiment of the invention uses two different flow paths for the coolant to cool the stator windings of the stator. The coolant can thus be conducted both via the recesses which accommodate the stator windings for direct cooling of the stator windings, and via separate channels for the coolant which are spaced apart from the recesses which accommodate the stator windings for indirect cooling of the stator windings.

(12) In order to define, determine or influence the flow path for the coolant through the recesses which accommodate the stator windings and/or through the channels which are configured in the stator laminated core of the stator at a distance to said recesses, the electric machine comprises the device by means of which the flow of coolant through the recesses and through the channels can be set depending on the operating point of the electric machine.

(13) At first operating points, the coolant flows only through the recesses which accommodate the stator windings. At the second operating points, the coolant can be conducted at least through the coolant channels. This allows particularly effective cooling of the stator of an electric machine and an advantageous increase in the continuous power density.

(14) The first operating points are preferably operating points at which an actual torque of the electric machine at an actual speed of the electric machine is greater than a threshold value. The second operating points are operating points at which the actual torque of the electric machine at the actual speed of the electric machine is less than the threshold value. Therefore, the first operating points are operating points at which the electric machine is operated at relatively high load. These are operating points at which the actual torque provided by the electric machine at the respective actual speed of the electric machine is greater than a threshold value, preferably greater than a defined percentage of the maximum torque that can be provided at said actual speed. The second operating points are operating points with relatively low load, i.e., operating points at which the actual torque provided by the electric machine at the actual speed of the electric machine is less than the threshold value.

(15) According to a first variant of an embodiment of the invention, at the second operating points,

a first portion of the coolant can be conducted through the channels and a second portion of the coolant can be conducted through the recesses, wherein the first portion of the coolant is greater than the second portion of the coolant. In this first variant, the device, via which the flow of coolant can be set depending on the operating point of the electric machine, preferably comprises an orifice ring which is capable of being driven in a rotating manner, which blocks the channels at the first operating points and unblocks the channels at the second operating points. The channels and the recesses then preferably use a common coolant inlet and preferably also a common coolant outlet.

(16) This first variant is structurally simple and therefore preferred. In this first variant it is further preferred that, regardless of the respective flow path used for the coolant, the coolant always flows around the winding heads of the stator windings.

(17) The orifice ring is preferably disposed at a coolant flow inlet-side end of the channels. Then, when the orifice ring is disposed at the flow inlet-side end of the channels and blocks the coolant channels at the first operating points, a pressure gradient at the orifice ring causes the channels blocked at the first operating points to be sealed, specifically by pressing the orifice ring against the flow inlet-side end of the stator laminated core.

(18) According to a second variant of an embodiment of the invention, at the second operating points, the coolant can be conducted only through the channels. In this second variant, the device, via which the flow of coolant can be set depending on the operating point of the electric machine, preferably comprises at least one valve which conducts the coolant only via the recesses at the first operating points and conducts the coolant only via the channels at the second operating points. The channels and the recesses then preferably use separate coolant inlets and preferably a common coolant outlet.

(19) The second variant also allows the flow of coolant to be set depending on the operating point. However, since the channels and the recesses use separate coolant inlets in this variant, this variant does not ensure flow around the winding heads of the stator windings at the coolant inlet side end of the stator laminated core at the second operating points.

(20) FIG. 1 shows a schematic cross-section through an electric machine **10**. The electric machine **10** comprises a rotor **11**, whereby a rotor shaft **12** and a rotor laminated core **13** of the rotor **11** are shown. The electric machine **10** further comprises a stator **14**. The stator **14** comprises a stator laminated core **15** and a housing **16**. Between the rotor **11** and the stator **14** there is an air gap **19**.

(21) The stator laminated core **15** of the stator **14** comprises recesses **17** in which stator windings **18** are disposed. FIGS. 2 and 3 show the stator windings **18** disposed in the recesses **17** for two of the recesses **17**. The stator windings **18** project from the stator laminated core **15** at axial ends of said stator laminated core and form winding heads **20** there.

(22) In the electric machine **10**, coolant can flow through the recesses **17** which accommodate the stator windings **18** in such a way that the stator windings **18** can be immediately and thus directly cooled by the coolant flowing through the recesses **17**. In FIGS. 1 and 2, the flow of coolant through the recesses **17** which accommodate the stator windings **18** is visualized by arrows **21**.

(23) To prevent the coolant flowing through the recesses **17** from flowing through the air gap **19** in the direction of the rotor **11**, a sleeve-like sealing element **22**, which is preferably fixedly connected radially on the inside to the stator laminated core **15**, is disposed in the air gap **19**. This sleeve-like sealing element **22** extends in axial direction beyond the axial ends of the stator laminated core **15** and consequently also covers the winding heads **20** with respect to the rotor **11**.

(24) In addition to the recesses **17** which accommodate the stator windings **18**, channels **23** are configured in the stator laminated core **15** at a distance to the recesses **17** which accommodate the stator windings **18**, whereby coolant can flow through these channels **23** as well. In FIGS. 1 and 2, the flow of coolant through the channels **23** is visualized by arrows **24**, whereby, in the shown design example, the channels **23** are configured in the stator laminated core **15** at a radial distance to the recesses **17**, specifically radially on the outside with respect to the recesses **17** in a so-called yoke of the stator laminated core **15**. Coolant, which flows through these channels **23** as shown by

arrows **24**, serves to cool the stator windings **18** indirectly. This coolant flowing through channels **23** therefore does not flow directly around the stator windings **18**.

(25) The electric machine **10** comprises a device **25** via which a flow of coolant through the recesses **17** which accommodate the stator windings **18** and through the channels **23** can be set depending on the operating point of the electric machine **10**.

(26) This device **25** can be used to set the flow of coolant in such a way that, at first operating points of the electric machine **10**, the coolant can be conducted and therefore flows only through the recesses **17** which accommodate the stator windings **18**, and, at second operating points of the electric machine **10**, the coolant can be conducted and therefore flows at least through the coolant channels **23**.

(27) The flow of coolant through the recesses **17** and channels **23** can therefore be set in a defined manner depending on the operating point of the electric machine **10**, in order to ensure particularly advantageous cooling of the stator **14** of the electric machine **10** and thus increase the continuous power density of the electric machine **10**.

(28) The first operating points of the electric machine **10**, in which the coolant flows only through the recesses **17** which accommodate the stator windings **18**, are operating points at which an actual torque of the electric machine **10** at an actual speed of the respective operating point of said electric machine is greater than a threshold value, preferably greater than a defined percentage of the maximum torque of the electric machine **10** that can be provided at the actual speed of the respective operating point.

(29) The second operating points of the electric machine **10**, in which the coolant can be conducted at least through the coolant channels **23**, are operating points of the electric machine **10** at which the actual torque of the electric machine **10** at the actual speed of the respective operating point of said electric machine is less than the threshold value.

(30) The threshold value can, for example, be 75% or 80% of the maximum torque that can be provided by the electric machine **10** at the actual speed of said electric machine.

(31) FIGS. **1** to **3** show a design example of an electric machine **10** in which, at the second operating points, a first portion of the coolant can be conducted through the channels **23** and a second portion of the coolant can be conducted through the recesses **17**, whereby the first portion of the coolant which flows through the channels **23** is greater than the second portion of the coolant which flows through the recesses **17**. In the design example of FIGS. **1** to **3**, the device **25**, via which the flow of coolant through the recesses **17** and the channels **23** can be set depending on the operating point of the electric machine **10**, comprises an orifice ring **26** which is capable of being driven in a rotating manner.

(32) At the first operating points of the electric machine **10**, the orifice ring **26** of the device **25** blocks the channels **23** so that the coolant then flows only through the recesses **17** which accommodate the stator windings **18** at the first operating points of the electric machine **10**. At the second operating points, the orifice ring **26** unblocks the channels **23**. At these second operating points, the coolant can then flow through both the channels **23** and the recesses **17**, whereby however a significantly greater portion of the coolant flows through the channels **23** at the second operating points as a result of the significantly lower flow resistance in the channels **23**.

(33) The orifice ring **26** is disposed at a coolant flow inlet-side end of the stator laminated core **15** and thus of the channels **23** configured in the stator laminated core **15**. Then, when the orifice ring **26** blocks the channels **23** at the first operating points, said ring is pressed against the end of the stator laminated core **15** as a result of the prevailing pressure gradient, which advantageously seals the channels **23**.

(34) In the design example of FIGS. **1** to **3**, the recesses **17** which accommodate the stator windings **18** and the channels **23** use both a common coolant inlet **27** and a common coolant outlet **28** configured on the housing **16** of the stator **14**.

(35) The design example of FIG. **1** to **3** has the advantage that, regardless of the operating point of

the electric machine **10** and thus regardless of the flow path used for the coolant, coolant can always flow around the winding heads **20** which project from the stator laminated core **15** at the axial ends thereof to cool said winding heads **20**. This applies in particular to the winding heads **20** which project from the recesses **17** at the coolant flow inlet-side end of the stator laminated core **18** around which cold coolant flows.

(36) FIG. **4** shows a design example of an electric machine **10** according to an embodiment of the invention, in which, at the second operating points of the electric machine **10**, the coolant can be conducted only through the channels **23** in the direction of flow **24**. In the design example of FIG. **4**, the device **25**, via which the flow of coolant can be set depending on the operating point of the electric machine **10**, comprises at least one valve **29a**, **29b**. At the first operating points, the valve **29a** is open and the valve **29b** is closed. At the second operating points of the electric machine **10** of FIG. **4**, the valve **29a** is closed and the valve **29b** is open. In FIG. **4**, therefore, at the first operating points, the coolant flows in the direction of flow **21** only through the recesses **17** which accommodate the stator windings **18** and, at the second operating points, the coolant flows in the direction of flow **24** only through the channels **23**.

(37) In the design example of FIG. **4**, channels **23** and the recesses **17** use separate coolant inlets **27a**, **27b**. The winding head **20** which projects from the recesses **17** at the coolant flow inlet-side end of the stator laminated core **15** can thus only be cooled in FIG. **4** when the coolant flows through the recesses **17** which accommodate the stator windings **18**. However, the winding head **20** which projects from the stator laminated core **15** at the coolant flow outlet-side end can have coolant flowing around it at all operating points.

(38) The electric machine **10** according to an embodiment of the invention thus uses a stator cooling system with different flow paths for the coolant, which are used depending on the operating point of the electric machine **10**. The flow of coolant is controlled or influenced via the device **25**, preferably via an orifice ring **26** which is capable of being moved in a rotating manner.

(39) At the first operating points of the electric machine **10** with a high load of the electric machine **10**, cooling takes place only via direct flow of the coolant around the stator windings **18**.

(40) At the second operating points of the electric machine **10** with low load, the coolant flows at least for the most part through the channels **23**. Less power is needed for a coolant pump conveying the coolant at the second operating points than at the first operating points, because the pressure loss in the channels **23** is less than in the recesses **17** which accommodate the stator windings **18**. The pump output can therefore be reduced at the second operating points and the range of an electric vehicle can thus be increased. Effective cooling at the first operating points can furthermore increase the continuous power density of the electric machine **10** at high load.

(41) In an embodiment, the invention further relates to a method for operating such an electric machine **10**. The method provides that, at the first operating points of the electric machine **10**, the coolant is conducted only through the recesses **17** which accommodate the stator windings **18** and, at the second operating points of the electric machine **10**, the coolant is conducted at least through the channels **23**.

(42) While subject matter of the present disclosure has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. Any statement made herein characterizing the invention is also to be considered illustrative or exemplary and not restrictive as the invention is defined by the claims. It will be understood that changes and modifications may be made, by those of ordinary skill in the art, within the scope of the following claims, which may include any combination of features from different embodiments described above.

(43) The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the

recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

Claims

1. An electric machine, comprising: a rotor comprising a rotor shaft and a rotor laminated core; a stator comprising a stator laminated core with recesses and stator windings disposed in the recesses of the stator laminated core, wherein coolant can flow through the recesses which accommodate the stator windings in such a way that the stator windings can be directly cooled by the coolant, wherein channels are configured in the stator laminated core at a distance to the recesses which accommodate the stator windings through which coolant can flow in such a way that the stator windings can be indirectly cooled by the coolant flowing through the channels; and a device via which a flow of coolant through the recesses which accommodate the stator windings and through the channels can be set depending on an operating point of the electric machine in such a way that, at first operating points, the coolant can be conducted only through the recesses which accommodate the stator windings and, at second operating points, the coolant can be conducted at least through the channels, wherein: the first operating points are operating points at which an actual torque of the electric machine at an actual speed of the electric machine is greater than a threshold value, and the second operating points are operating points at which the actual torque of the electric machine at the actual speed of the electric machine is less than the threshold value.
2. The electric machine according to claim 1, wherein, at the second operating points, a first portion of the coolant can be conducted through the channels and a second portion of the coolant can be conducted through the recesses, wherein the first portion of the coolant is greater than the second portion of the coolant.
3. The electric machine according to claim 1, wherein the device, via which the flow of coolant can be set depending on the operating point of the electric machine, comprises an orifice ring which is capable of being driven in a rotating manner, and the orifice ring blocks the channels at the first operating points and unblocks the channels at the second operating points.
4. The electric machine according to claim 3, wherein the orifice ring is disposed at a flow inlet-side end of the stator laminated core and thus of the channels.
5. The electric machine according to claim 4, wherein the channels and the recesses use a common coolant inlet.
6. The electric machine according to claim 5, wherein the channels and the recesses also use a common coolant outlet.
7. The electric machine according to claim 1, wherein, at the second operating points, the coolant can be conducted only through the channels.
8. The electric machine according to claim 7, wherein the device, via which the flow of coolant can be set depending on the operating point of the electric machine, comprises at least one valve, and the at least one valve conducts the coolant only via the recesses at the first operating points and conducts the coolant only via the channels at the second operating points.
9. The electric machine according to claim 1, wherein the channels and the recesses use separate coolant inlets.
10. The electric machine according to claim 9, wherein the channel and the recesses also use a common coolant outlet.

11. A method for operating the electric machine configured according to claim 1, wherein, at the first operating points, the coolant is conducted only through the recesses, and at the second operating points, the coolant is conducted at least through the channels.

12. An electric machine, comprising: a rotor comprising a rotor shaft and a rotor laminated core; a stator comprising a stator laminated core with recesses and stator windings disposed in the recesses of the stator laminated core, wherein coolant can flow through the recesses which accommodate the stator windings in such a way that the stator windings can be directly cooled by the coolant, wherein channels are configured in the stator laminated core at a distance to the recesses which accommodate the stator windings through which coolant can flow in such a way that the stator windings can be indirectly cooled by the coolant flowing through the channels; and a device via which a flow of coolant through the recesses which accommodate the stator windings and through the channels can be set depending on an operating point of the electric machine in such a way that, at first operating points, the coolant can be conducted only through the recesses which accommodate the stator windings and, at second operating points, the coolant can be conducted at least through the channels, wherein, at the second operating points, a first portion of the coolant can be conducted through the channels and a second portion of the coolant can be conducted through the recesses, and wherein the first portion of the coolant is greater than the second portion of the coolant.

13. The electric machine according to claim 12, wherein the first operating points are operating points at which an actual torque of the electric machine at an actual speed of the electric machine is greater than a threshold value, and the second operating points are operating points at which the actual torque of the electric machine at the actual speed of the electric machine is less than the threshold value.

14. The electric machine according to claim 12, wherein the device, via which the flow of coolant can be set depending on the operating point of the electric machine, comprises an orifice ring which is capable of being driven in a rotating manner, and the orifice ring blocks the channels at the first operating points and unblocks the channels at the second operating points.

15. The electric machine according to claim 12, wherein the channels and the recesses use separate coolant inlets.

16. An electric machine, comprising: a rotor comprising a rotor shaft and a rotor laminated core; a stator comprising a stator laminated core with recesses and stator windings disposed in the recesses of the stator laminated core, wherein coolant can flow through the recesses which accommodate the stator windings in such a way that the stator windings can be directly cooled by the coolant, wherein channels are configured in the stator laminated core at a distance to the recesses which accommodate the stator windings through which coolant can flow in such a way that the stator windings can be indirectly cooled by the coolant flowing through the channels; and a device via which a flow of coolant through the recesses which accommodate the stator windings and through the channels can be set depending on an operating point of the electric machine in such a way that, at first operating points, the coolant can be conducted only through the recesses which accommodate the stator windings and, at second operating points, the coolant can be conducted at least through the channels, wherein: the device, via which the flow of coolant can be set depending on the operating point of the electric machine, comprises an orifice ring which is capable of being driven in a rotating manner, and the orifice ring blocks the channels at the first operating points and unblocks the channels at the second operating points.

17. The electric machine according to claim 16, wherein the first operating points are operating points at which an actual torque of the electric machine at an actual speed of the electric machine is greater than a threshold value, and the second operating points are operating points at which the actual torque of the electric machine at the actual speed of the electric machine is less than the threshold value.

18. The electric machine according to claim 16, wherein, at the second operating points, a first

portion of the coolant can be conducted through the channels and a second portion of the coolant can be conducted through the recesses, wherein the first portion of the coolant is greater than the second portion of the coolant.

19. The electric machine according to claim 16, wherein the orifice ring is disposed at a flow inlet-side end of the stator laminated core and thus of the channels.

20. The electric machine according to claim 19, wherein the channels and the recesses use a common coolant inlet.
